# ASSESSMENT - 3

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# Descriptive written Test

1. Intelligent Agents in a Smart Home Automation system

Introduction:

An intelligent agent is a system that perceives its environment through sensors and take actions using actuators to achieve specific objectivies. In a smart home, agents control lighting, temperature, and security basedon motion, environmental conditions, time, and user preference.

Types of Intelligent Agent

1. Simple Reflex Agent:

→ work based on the current input using condition - action rules.

→ Dose not store past experencio on remembes previous states.

Ex:- The motion detected lights turn on, off.

2. Model-Based Agent:

* Maintain an intemal model of the environment and uses past information to make decision.

* Ex:- Remembers that room is occupied even when the motion sensor temporarily stops detecting movment.

Advantages:- works effectively in partially observable environments.

3. Goal-Based Agent:-

- select action to achieve specific Goals

Ex:- It adjust the room temperature to reach the user's Prefered temp and activates security system when the house is unoccupied.

Advantages :- can plan actions to achieve desired outcomes.

Disadvantages :- requires planning and may take more time to make decision.

4. Utility-Based Agent:-

⇒ chooses actions by Evaluating their usefulness based on factors such as Comfort, energy Consumption cost and the security.

Ex:- It balances air-conditioning Comfort with electricity and savings while maintaining a safe indoor temperature.

Advantages: Helps optimize multiple objectives simultaneously.

Disadvantages:

Designing an appropriate utility function can be Complex.

Most Effective Approach:-

A hybrid approach Combining model-based and utility-based agents, support by goal-based planning and simple reflex rules, would be most effective.

Conclusion:-

A hybrid intelligent agent is the best choice that because it combines fast reactions, environmental awareness, planning, and optimization. It can improve comfort, reduce energy consumption and provide effective home security.

## 2. Solving an Unknown Maze using UCS and IDS

Introduction:

A robat navigating an unknown maze that must find a path from its starting position to the exit without knowing the complete layout beforehead. uninformed search algorithm explore possible paths without using heuristic information about the distance to the goal.

1. Uniform Cost search (UCS):

uniform cost search expands the node with the lowest total path cost from the starting point.

working:

1. place the starting point in a priority quee with zero cost.
2. Select the position with the lowest accumlated path cost.
3. Explore its avaliable neighboring positions.
4. update the path cost add unexplored positions to the queue.
5. Contine until the exit is reached.

Ex:- if one path cost 10 units, anoth 6 units select which has low cost.

Advantages:-

* Finds the least-cost solution when step costs are non-negaitive.

Disadvantages:-

* May explore many positions before reaching the exit.

2. Iterative Deepening search (IDS) :-

IDS repeadly perform depth-limited, depth-first Seach, incresing the depth limit in each iteration.

working:

1. Start with a depth limit of zero.
2. Seach all positions within the current depth limit
3. If the exit is not fount, increse the limit by one.
4. Repeat the until the exit found.

Ex:- The Robot first seaches depth 0, thed depth 1, depth 2 and so on until it reaches the exit.

Advantages:-

* use less memory than UCS
* Is complete when the branching factor is finite.

Disadvantages:-

* Repeats the exploration of nodes at shallow depths.

Most Effective Combination

A model-based, goal-based robot using UCS is suitable when different paths have different movement cost and finding the least-cost route is important.

Conclusion:-

UCS is prefeable when the robot must find the least-cost path, while IDS is preferable when memory is limited and all movements have equal cost.

Combining a model-based agent with goal-based agent planning improves the robot's ability to remember explored ares, make informed decision, and reach the exit efficiently.